

Three layers of realisation: hierarchy recovery, holonomy, information and contextuality

Abstract

This paper consolidates three bodies of work into one account. Each result carries its source and its status: proved, computed, measured, open, or closed by theorem.

1. *Hierarchy recovery* [H]. A gradient-free estimator (spectral initialisation, soft hand-off, inside-outside EM and split-merge) recovers a depth-4 probabilistic grammar to 0.001 nats per sequence of the true model in 131 s. Twelve transformer configurations never represent the level-3 latent symbol.
2. *Cohomological strand* [O]. Holonomy of known operations gives a genuine obstruction $[g] \in H^1(\Gamma, A)$. Marginal obstructions within one data source vanish (T0). The information site and the contextuality site are separated by a theorem.
3. *Results of the present work* [S]:
 - an exact identifiability criterion for holonomy estimated from data, with a calibrated test and its sample cost;
 - a direct construction from operations to outcomes, $[g] \neq 0 \iff \gamma \neq 0$, which is invisible to the probability layer;
 - for lossy operations: a closed form $\text{CF} = \max(0, 1 - |L|\varepsilon/2)$ on a cycle, and a holonomy-girth lower bound on general graphs;
 - the loop spectrum as a monoid-valued holonomy invariant;
 - a decorated Bregman–Čech nerve that carries orientation and phase, with a codimension criterion for when it can;
 - restriction and extension maps between the information and contextuality views;
 - a gated contextuality test for streams with *not-applicable* entries, with a pinned maximum-likelihood null.

The organising statement is that the three layers (operations, probability and outcomes) respond to the same data through different invariants, and that no invariant of one layer determines another.

Provenance tags.

- [H]: *Recovering latent hierarchy without gradient descent* (hierarchy paper and handoff).
- [O]: *Holonomy obstructions for composed local operations* (holonomy paper).
- [S]: this work, implemented in the `amb.vigneaux` package. It has 107 tests, and every numerical claim tagged [S] is produced by that code.

Status tags: proved, computed (exact computation or verified identity), measured (empirical, no proof), open, closed (settled by theorem, not to be reopened).

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1 The standing constraint

Theorem 1.1 (T0: single-source global section). *[O, Thm 7] proved Let $\{P_C\}_{C \in \mathcal{M}}$ be context marginals estimated from one sample $\mathcal{D}$. The empirical joint $P_{\mathcal{D}}$ restricts to each P_C , so the family extends. Consequently the Čech obstruction $\gamma(s)$ vanishes for every section and the contextual fraction is 0, on any cover, acyclic or not.*

Proof. $P_{\mathcal{D}}$ is a global section by construction. □

Corollary 1.2. *[H/O] proved A single fitted model can never exhibit an obstruction to its own consistency. If an estimator and a detector share a data source, the detector measures model misfit, not structure.*

Remark 1.3 (T0 as a functor). [S] proved Restriction R from global laws to context families (Section 20) has image exactly the non-contextual models. So T0 is the statement that anything in the image of R has $\text{CF} = 0$ and $\gamma = 0$. The engine’s first version violated T0 by resampling a single-source world context by context: it produced a spurious CF up to 0.10, and the fitted model’s CF equalled its signalling defect exactly. After adding a single-source sampling mode, $\text{CF}(\hat{P}_N) \leq 2 \times 10^{-16}$.

2 The three layers

Layer	Invariant	Condition for non-triviality
1 Operations	$H^1(\Gamma, A) \cong A^{\beta_1}$, holonomy; the loop spectrum for lossy operations	The operations are known or identifiable (Prop. 15.1); $\beta_1 > 0$
2 Probability	$H^1(\mathcal{S}; \mathcal{F}_\alpha)$, one-dimensional, generated by entropy (Tsallis for the α -action)	The conditioning action is present; $I = \delta_t H$ measures its failure to be trivial
3 Outcomes	$\gamma \in \check{H}^1(\mathcal{U}, \mathcal{F}_{\tilde{C}})$; CF	The cover has no global context; data come from more than one source (T0)

Table 1: The stratification [O, §6], extended [S].

Remark 2.1 (Stratification, not bridge). [O, Rem. 18] proved Each invariant is non-trivial on its own layer and zero on the others, and the reasons are the hypotheses themselves. Layer 2’s site has a global joint, which gives the observable monoid an absorbing element; Theorem 13.3 then kills every trivially-acting coefficient system, which is where Layer 3’s coefficients would have to live. Layer 3’s site is built so that no global context exists. That is what permits $\gamma \neq 0$, and it simultaneously denies Layer 2 the joint it requires.

[S] adds that the layers are connected *around* Layer 2, not through it. There is an exact construction from Layer 1 to Layer 3 (Theorem 16.1), and Layer 2 is provably blind to it (Proposition 16.2).

Part I

Hierarchy recovery [H]

3 Setup and exact floors

All experiments use the Random Hierarchy Model of Cagnetta et al. [9]. It is a probabilistic context-free grammar on a fixed complete binary tree of depth L , with V symbols per internal level, m productions per symbol, and a leaf alphabet of size A . Because the tree shape is fixed and positional, the latent structure is a Markov chain from leaves to level 1 and on up to the root.

- *Corpus A*: $L = 4$, $V = 64$, $A = 48$, $m = 12$, 2×10^5 sequences of length 16.
- *Corpus B*: $L = 4$, $V = 8$, $A = 12$, $m = 3$, 2×10^5 sequences.

The exact conditional entropy $H(x_t \mid x_{<t})$ is computed by inside–outside (Theorem 10.1). For Corpus A: $H_{\text{local}} = 1.6034$, $H_{\text{higher}} = 3.5527$, $H(x_0) = 3.8161$ nats. The latent-symbol entropies given the prefix, for levels 1–4, are 2.7835, 2.2923, 2.2900, 2.7910 against chance $\ln V = 4.1589$. About 1.87 nats of level-3 information are recoverable from the prefix.

4 Transformers do not recover levels 3–4

configuration	local	higher	level-3 probe
2 layers, $d = 64$ (baseline)	2.173	3.831	0.028
4 layers	2.078	3.797	—
8 layers, frozen embedding	2.037	3.755	—
8 layers, frozen emb., fresh sampling	2.022	3.772	0.027
$d = 128$, 4 layers	2.039	3.770	—
frozen random embedding (2 layers)	2.186	3.824	—
constructed order coordinate	2.162	3.815	—
auxiliary supervision, levels 3–4	2.088	3.777	0.056
latent marginalising head (stopped at 34k)	3.227	3.835	—
dilated causal CNN	2.104	3.827	0.032
curriculum over tree depth	2.085	3.803	0.034
curriculum control (no easy cases)	2.043	3.789	0.033
exact floor	1.603	3.553	—

Table 2: Corpus A, 48,000 steps each. Shuffled-label controls read 0.021–0.034; chance is $1/V = 0.0156$. [H]

Statement [H] *measured*. Across twelve configurations the level-3 probe never leaves the shuffled-label floor, although 1.87 nats of level-3 information are present in the prefix. A linear probe on the embedding was *withdrawn*: a random matrix supports the same probe accuracy.

Hypothesis (not established). Recovering the level- ℓ symbol requires marginalising over the parses consistent with the prefix. A feedforward stack has only an upward pass, while inside–outside couples an upward and a downward pass.

5 The estimator and its verification

Construction 5.1 (Estimator). [H]

1. *Spectral initialisation*: build the pair-context co-occurrence matrix, take its leading singular directions, and k -means the pairs into V groups.
2. *Soft hand-off*: pass the posterior to the next level, $q_\ell \propto \sum_{x,y} P_\ell[\cdot, x, y] q_{\ell-1}^{\text{lo}}(x) q_{\ell-1}^{\text{hi}}(y)$.
3. *EM*: inside-outside E-step and normalised-counts M-step, keeping the sweep with the best held-out likelihood.

check	statement	max error
V0	EM never decreases the training log-likelihood	$+1.8 \times 10^{-2}$ (min step)
V1a	a free subtree has inside value exactly 1	1.8×10^{-15}
V1b	restriction = marginalisation	8.9×10^{-16}
V1c	gluing = independence join, $P(x_{0:16}) = \sum_{u,v} \mu(u, v) \beta_L(u) \beta_R(v)$	1.0×10^{-14}
V2	the soft hand-off is a sufficient statistic	6.7×10^{-16}
V4a	merge cost = $N I(\text{children}; B \mid q(B))$	1.2×10^{-14}

Table 3: Identities of the tree model class [H] computed. They verify the implementation and that the fitted object is a sheaf section. They do not test whether the data are tree-generated. The hard (argmax) decode flips on 34–46% of sequences under a chain of conditionings; the soft decode does not change.

variant (Corpus A, 2×10^4 sequences)	local	higher	held-out log p/seq
joint EM from random init	2.566	3.831	−51.18
bottom-up, frequency-rank clusters	2.598	3.833	−51.45
spectral + soft hand-off + EM (10 sweeps)	2.227	3.802	−48.23
spectral + soft hand-off + EM (50 sweeps)	2.162	3.807	−47.75
stagewise + vertical re-clustering	2.240	3.802	−48.34
transformer, 48k steps	2.022	3.772	—
true grammar	1.599	3.553	−41.21

Table 4: Estimator variants on Corpus A [H].

6 Instruments for detecting hierarchy

- **V6, leaf pairs** [H] measured. The test is a parametric bootstrap of empirical mutual information against datasets sampled from the fitted model.
 - *Planted positive*: copying leaf 7 into leaf 8 with probability 0.3 gives $z = +220$, against +6.5 elsewhere; even the true grammar is refuted there ($z = +390$).
 - *Limitation*: on Corpus A, pairwise leaf MI across level-2 and root boundaries is about 10^{-4} even under the true grammar, while the plug-in bias floor is about 0.055. So V6 has no power above level 1.
- **V7, blocks** [H] measured. Each block is decoded by its own inside vector, and adjacent blocks are compared under the disjointness invariant.

- Decoded with the true grammar, blocks sharing a parent carry about 1 nat, and others 0.002–0.014.
- The fitted model captures 79%, 1.4% and 3.9% of this at levels 1–3.
- β_1 of the block complex [H] measured. A tree predicts $\beta_1 = 0$ at every threshold.
 - Corpus B, unplanted: 0 at all 25 thresholds.
 - Two planted block couplings: $\beta_1 = 1$ at 5 thresholds, with the null at 0. A single coupling never produces a cycle.
 - β_1 is computed on a Rips complex, so it is a calibrated statistic and not an approximation to Čech persistence (Theorem 10.20).

7 The regime determines learnability

	level 1	level 2	level 3	level 4
Corpus A: observations per parameter	1.4	0.31	0.15	0.08
Corpus B: observations per parameter	1389	1563	781	391

Table 5: Identifiability regime [H]. Levels fail at 0.15 observations per parameter and succeed at 781.

On Corpus B at 200 sweeps, the level-3 block dependence captured rises from 0.0854 to 0.1827, and its discrepancy falls from $z = +20.2$ to $+5.4$. The earlier failure was therefore an unestimable regime, then undertraining. Merges at levels 1–3 cost real conditional MI; the root alphabet is redundant, with $8 \rightarrow 2$ improving held-out likelihood by 0.004.

8 Closing the gap: basin selection and split–merge

start	held-out log p /seq	local / higher
exact truth ($\varepsilon = 0$)	−15.324	0.3322 / 1.5833
truth + 50% / 80% / 90% uniform	−15.325 / −15.325 / −15.330	—
truth + 95% / 99% uniform	−15.775 / −18.113	—
uniform	−20.706	0.5091 / 2.0791
spectral init (ours)	−17.511	0.3952 / 1.7937

Table 6: Diagnosis: EM started at the truth [H] measured. The truth is a fixed point with a very wide basin.

A hard partition of observed child pairs cannot represent a grammar in which pairs are shared across parents (4 of 16 at level 1).

- Sharpening the clustering hurts: sibling context gives −18.06.
- Blurring it helps: blending with uniform at weight 0.9 gives −15.87.

Construction 8.1 (Split–merge). [H] For each level:

1. duplicate the symbols with a perturbation;

2. run EM;
3. greedily merge back by Bregman information over *all* $2V$ symbols, so merges can recombine across the split;
4. accept only if held-out likelihood improves.

stage	held-out $\log p/\text{seq}$	merge cost (nats/seq), levels 1–4
warmup, 40 EM sweeps	−17.187	—
round 1	−15.491	0.160, 0.113, 0.008, 0.003
round 2	−15.325	0.042, 0.011, 0.007, 0.005
round 3	−15.3253	0.052, 0.020, 0.007, 0.004
true grammar	−15.3243	—

Table 7: Split–merge on Corpus B [H] *measured*: 131s, local/higher 0.3322/1.5835 against a true 0.3322/1.5833.

The recovered model passes all four instruments:

- V3 matches the true column at every boundary.
- V6 flags 0 of 18 pairs, with largest $|z| = 1.7$.
- V7 at level 3 reads 0.5863/0.5882 against a true 0.6140.
- β_1 against the fitted null is 0 of 25, where the local-optimum fit gave 6 of 25.

9 Why clustering is necessary at scale

A dense inside pass costs about $2NV^3$. At $N = 10^6$ and $V = 1024$ a sweep costs about 4.3×10^{15} . Compression to $K = 16$ gives about 8×10^{11} over 50 sweeps, a speedup of $(V/K)^3 = 262,144$. For m -ary grammars the order rises to $m + 1$, so compression buys $\log_K V$ orders of headroom.

Compression is not free. Merge costs are level-dependent and must be measured, not assumed; V4a supplies the measurement. At $L = 16$ the root is observed only about 15 times per 10^6 tokens, so compression removes the compute barrier but not the identifiability barrier.

10 Theorems and constructions used [H]

Theorem 10.1 (Inside–outside; Baker, Lari–Young). *used* For a PCFG on a fixed tree, the inside values β and outside values α give exact marginals: the posterior at a node is $\propto \alpha\beta$.

Theorem 10.2 (EM ascent; Dempster–Laird–Rubin). *used* Each EM iteration does not decrease the observed-data log-likelihood.

Theorem 10.3 (Convergence; Wu). *used* Under regularity conditions, EM converges to a stationary point, and the limit depends on the initialisation.

Theorem 10.4 (Extension; Vorob’ev). *used* Every consistent family of marginals on a hypergraph extends to a global distribution if and only if the hypergraph is acyclic. A fixed tree is acyclic.

Theorem 10.5 (Independent pullback; Simpson Thm 6.4, Prop. 8.5). *used* A presheaf on standard Borel probability spaces is an atomic sheaf iff it maps independent squares to pullbacks. The independent pullback is the conditional product.

Definition 10.6 (Supports; Simpson Def. 7.1). A support of a random variable is its terminal factorisation through a sample space.

Theorem 10.7 (Conditional products; Fritz §12). *used In a Markov category with conditionals, conditional products exist and satisfy the Dawid–Studený axioms. FinStoch is such a category.*

Theorem 10.8 (Sufficient statistics; Fritz Def. 14.3, Thm 14.5). *used A statistic is a deterministic morphism, and the abstract Fisher–Neyman factorisation holds. The soft hand-off is a statistic, and $V2$ verifies its sufficiency.*

Theorem 10.9 (Minimal sufficiency; Fritz Def. 16.2, Bahadur). *used A minimal sufficient statistic is least informative among the sufficient ones; a complete sufficient statistic is minimal.*

Remark 10.10 (Fritz and cohomology). Fritz’s framework supplies semantics for conditional independence and sufficiency, but no cohomology. It does not bear on a map from information cohomology into a persistence module.

Construction 10.11 (Independence join; Wright §7.1). For tables agreeing on a shared marginal, $(p \bowtie q)_{ijk} = p_{ij}q_{jk}/r_j$ is the maximum-entropy weak pullback.

Construction 10.12 (Merging as convex combination; Wright §7.2). Merged rows are mixtures, averaged in the Giry monad.

Theorem 10.13 (Bregman information; Bregman, Banerjee et al.). *used For KL , the optimal cluster representative is the mean, and the clustering loss equals the mutual information between the data and the cluster assignment.*

Construction 10.14 (Merge criterion). Merging symbols costs

$$\sum_s n_s D_{\text{KL}}(R_s \| R_{\text{class}}) = N I(\text{children}; B \mid q(B)).$$

The cost is zero iff the merge is lossless, which is strong lumpability (Kemeny–Snell).

Construction 10.15 (Agglomerative merging; Slonim–Tishby). Greedy least-loss merging is the agglomerative information bottleneck; split–merge (Petrov et al.) is its grammar analogue.

Theorem 10.16 (Ostrowski). *used Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to the real absolute value or a p -adic one, and the resulting completions are pairwise inequivalent.*

Construction 10.17 (Bias correction; Miller–Madow). Plug-in MI on a $K \times L$ table is biased upward by about $(K - 1)(L - 1)/2N$.

Construction 10.18 (β_1). $\beta_1 = \dim \ker \partial_1 - \text{rank } \partial_2$, computed over $\text{GF}(2)$ on the 2-skeleton.

Proposition 10.19 (The lattice quasi-morphism does not exist). *[H, Prop. 1] proved, closed Let $\rho : \mathbb{Q} \rightarrow L$ be order-preserving for the real order and constant on each p -adic ball. Then ρ is constant.*

Proof. A p -adic ball is dense in $\mathbb{R}$. Choose $x < y < z$ with x, z in one ball B_1 and y in another ball B_2 . Monotonicity gives $c_1 \leq c_2 \leq c_1$. $\square$

Theorem 10.20 (Bregman TDA; Edelsbrunner–Wagner). *used KL restricted to the simplex is of Legendre type. Čech and Delaunay complexes of Bregman balls have the homotopy type of the union (Contractibility Lemma). The Rips filtration can be arbitrarily far from Čech (Result 2). Measured $[H]$: maximum β_1 is 20–21 for Čech and 0 for Rips.*

Refuted, do not retry [H]:

- sibling-only clustering context (-18.06);
- deterministic annealing ($+0.22$ nats, independent of T_0);
- stagewise plus vertical re-clustering (-48.34 against -47.75);
- posterior entropy as a confidence criterion: posterior entropies are 0.198 – 1.003 at levels capturing under 4% , whereas $I(\text{parent}; \text{children})$ reads $10.1, 8.1, 4.5, 5.3$ against a true 8.9 – 9.0 ;
- curriculum transfer: the control gives $2.043/3.789$ against $2.085/3.803$.

Part II

Holonomy and the separation of sites [O]

11 The holonomy obstruction

Definition 11.1 (Operation system). [O, Def. 1] (Γ, A, g) consists of a finite connected graph $\Gamma = (V, E)$, an abelian group A , and $g : E \rightarrow A$ with $g_{ba} = -g_{ab}$.

Definition 11.2 (Global field). [O, Def. 2] A map $c : V \rightarrow A$ with $c(b) - c(a) = g_{ab}$ for every edge.

Theorem 11.3 (Holonomy obstruction). [O, Thm 3] *proved, computed* A global field exists iff the holonomy $h(L) = \sum_{e \in L} \pm g_e$ vanishes on every cycle. The obstruction is $[g] \in H^1(\Gamma, A) \cong A^{\beta_1}$, with $\beta_1 = |E| - |V| + 1$.

Proof. Integrate g along a spanning tree. A non-tree edge is consistent iff its fundamental cycle has zero holonomy, and the fundamental cycles form a basis of the cycle space. $\square$

Verified against brute force on a room with $|V| = 4$, $|E| = 5$, $\beta_1 = 2$, $A = \mathbb{Z}/4096$. The holonomy is $(0, 0)$ when a field exists, and $(0, 1)$ and $(2, 1)$ when none does [O]. The same computation in [S] gives the same classes, up to a unimodular change of cycle basis.

Theorem 11.4 (Holonomy as a connecting class). [O, Thm 4] *proved* Consider the exponential sequence $0 \rightarrow \mathbb{Z}_M \rightarrow \mathbb{R}_M \rightarrow \mathcal{S}_M^1 \rightarrow 0$ on a manifold M with boundary. Since $\mathbb{R}_M$ is soft, the connecting map $\delta_M : H^0(M, \mathcal{S}^1) \rightarrow H^1(M, \mathbb{Z})$ is an isomorphism modulo global real lifts. Naturality under $r : \partial M \hookrightarrow M$ gives $r^*(\delta_M[g]) = \delta_{\partial M}(r^*[g]) = \text{Hol}(\alpha, \partial M)$.

Remark 11.5 (What Theorem 11.4 does not say). [O, Rem. 5] $\delta_M[g]$ lies in $H^1(M, \mathbb{Z})$, not in $\text{Ext}^1(\mathcal{S}_M^1, \mathbb{Z}_M)$. It is not the AMB obstruction: the “support reduction” $\mathcal{F}_{\text{prob}} \rightarrow \mathbb{Z}$ is not additive, so it is not a morphism of sheaves.

Proposition 11.6 (T0 does not apply to operation systems). [O, Prop. 8] *An operation system has no sample; a witness would be a global field, which is the object in question.*

Proposition 11.7 (Vorob’ev does not apply). [O, Prop. 9] *Here the relevant structure is the cycle space of Γ : $\beta_1 = 0$ makes H^1 trivial, while $\beta_1 > 0$ leaves it free.*

Proposition 11.8 (Possibilistic fragility does not apply). [O, Prop. 10] *computed* The AMB obstruction is a functional of the support, so smoothing kills it. Measured: γ falls from $9/9$ to $0/27$ at $\varepsilon = 10^{-9}$, while $\text{CF} = 1 - 3\varepsilon$ survives to $\varepsilon = 1/3$. The holonomy class is a sum of the operations themselves and has no support to blur.

attempted coefficients	why it failed	verified
p -adic lattice $\rho : \mathbb{Q} \rightarrow L$	order-preserving and constant on balls implies constant	proof
probability simplex Δ^{V-1}	a convex set with no inverses, so no δ	—
support indicator $\omega = \int_{\text{supp}} \mathbf{1} dP$	identically 1 on every face of every family	computed
rounding $\mathbb{R} \rightarrow \mathbb{Z}$	not additive, so not a module map	—
$\mathbb{Z}/n$ on edges	abelian, δ exists, the class varies	computed

Table 8: The coefficient group is what mattered [O, §3].

Remark 11.9 (β_1 changes role). [O, Rem. 11] In the marginal setting, β_1 of the incidence complex is a function of the cover alone. It equals 4 for the obstructed 3×3 parity family and for both controls, so it cannot map to γ . In an operation system, β_1 is the rank of the obstruction group.

12 The two coboundary operators

Definition 12.1 (Conditioning action). [O, Def. 12] $(S_0 \cdot F)(P) = \sum_j P(S_0 = v_j) F(P \mid S_0 = v_j)$.

Definition 12.2 (The two coboundaries). [O, Def. 13]

$$(\delta F)(S_0; \dots; S_k) = S_0 \cdot F(S_1; \dots; S_k) + \sum_{i=1}^k (-1)^i F(\dots; S_{i-1} S_i; \dots) + (-1)^{k+1} F(S_0; \dots; S_{k-1}).$$

δ_t is the same expression with the first term replaced by $F(S_1; \dots; S_k)$. Both operators square to zero.

Theorem 12.3 (Baudot–Bennequin). [O, Thm 14]; [5] used; verified numerically [S]

1. $\delta c = 0$, so $H^0 \cong \mathbb{R}$.
2. $\delta H = 0$ (the chain rule), and under locality hypotheses $H^1(\mathcal{F}, \delta)$ is one-dimensional, generated by H .
3. $\delta_t H = I$, and $(\delta - \delta_t)H = (S_0 - 1) \cdot H(S_1) = -I$.

[S] verifies numerically: $\delta H = 0$ and $\delta_t H = I$ to 10^{-10} ; $(XY) \cdot F = X \cdot (Y \cdot F)$; $\delta^2 = 0$; and Rényi-2 entropy is not a cocycle.

Remark 12.4 (Easy to get backwards). [O, Rem. 15] $I_2 = \delta_t H$, not δH . Higher I_k are coboundaries for δ or for δ_t according as k is odd or even. No regularity or symmetry condition separates H from I .

13 Non-unification and vanishing

Theorem 13.1 (Non-unification). [O, Thm 16] proved No morphism of coefficient systems carries a nonzero AMB class in $\mathcal{F}_{\mathbb{Z}}$ to a nonzero class in $\mathcal{F}_{\alpha}$.

Remark 13.2 (Holonomy is not the bridge). [O, Rem. 17] The holonomy class shares its coefficient type with AMB and its locality with Vigneaux, but it is a cocycle of transformations, not of sections or information functions. So it is a third object.

Theorem 13.3 (Vanishing on the trivially-acting complex). *[O, Thm 27] proved, computed* Let $\mathcal{S}$ be an information structure whose observables form a monoid under joint, with an absorbing element, and let A be an abelian group with trivial action. Then $H^n(\mathcal{S}; A) = 0$ for $n > 0$ and $H^0 = A$.

Proof. With trivial action the cochains are constants, and the complex is the bar complex of a monoid with an absorbing element, whose classifying space is contractible. $\square$

Verified for the join-semilattices of two and three binary variables: $H^0 = \mathbb{Z}$ and $H^1 = H^2 = 0$ over $\mathbb{Q}$, and over $\mathbb{Z}$ by Smith normal form, with no torsion.

Corollary 13.4 (Extensions do not help). *[O, Cor. 28] proved, closed* The connecting map of any extension of $\mathcal{F}_\alpha$ by a trivially-acting $\mathcal{F}_\mathbb{Z}$ lands in $H^{k+1}(\mathcal{F}_\mathbb{Z}) = 0$.

Remark 13.5 (The dichotomy). [O, Rem. 29] Either the coefficients act trivially, and Theorem 13.3 applies; or they do not, and the AMB class does not live there. An information structure presupposes that every set of variables has a joint. A measurement scenario presupposes that some do not.

Remark 13.6 (Scope of Theorem 13.3). [S] The theorem concerns the *trivially-acting* complex. Under the non-trivial conditioning action, H^1 is one-dimensional (Theorem 12.3); under the α -deformed action it is generated by Tsallis entropy (Section 18). $H^{\geq 2}$ is open for both actions (Baudot–Bennequin, Problem 1). The finite linear algebra of `h2_rational.py` does *not* transfer to the deformed action: cochains are then functions of P , and the spaces are infinite-dimensional without a restricted function class.

Problem 13.7 (Site correspondence). [O, Prob. 21] *open* Does an AMB scenario determine an information structure in Vigneaux’s sense?

Problem 13.8 (Cover preservation). [O, Prob. 22] *open* Is the correspondence a morphism of sites? This depends on Problem 21.

Problem 13.9 (Coefficient comparison). [O, Prob. 23] *blocked* A morphism $\Psi : \mathcal{F}_\mathbb{Z} \rightarrow \mathcal{F}_\alpha$ intertwining the module structures has an undefined left-hand side.

Problem 13.10 (Extension formulation). [O, Prob. 24] *settled negatively* by Cor. 28

Problem 13.11 (Parity criterion). [O, Prob. 25] A mod- p family is contradictory iff every variable lies in an even number of contexts and $\sum_C s_C \not\equiv 0 \pmod{p}$. Brute force caught two failed examples. [S] reproduces the mod-3 grid exactly (Section 14).

Problem 13.12 (Holonomy from estimated operations). [O, Prob. 26] *answered in part* by §15

Closed by theorem, not to be reopened [O]:

- the p -adic bridge (Prop. 10.19, Theorem 10.16);
- branch cohomology with simplex coefficients: a rooted tree poset has an extremal object, so $H^{k>0} = 0$ for every coefficient system;
- a map from β_1 of the incidence complex to γ ;
- a class for $I(X; Y \mid Z)$ obtained by regularity restrictions;
- the support-reduction identification of boundary holonomy with γ .

Part III

Results of the present work [5]

14 Engine and verification of the outcome layer

The `amb_vigneaux` package implements three things:

- the outcome layer: support presheaves, logical and strong contextuality, and γ decided exactly over $\mathbb{Z}$ (by integer diagonalisation), $\mathbb{Q}$ and $\mathbb{Z}/p$;
- the contextual fraction by LP, with its dual Bell inequality;
- the probability layer: conditioning action, δ , δ_t and co-information.

It reproduces the standard values computed:

- PR box and GHZ: strongly contextual, $\text{CF} = 1$, $\gamma \neq 0$ at every section.
- Tsirelson box: $\text{CF} = \sqrt{2} - 1$; noisy Tsirelson: $\text{CF} = \max(0, \sqrt{2}v - 1)$; noisy PR box: $\text{CF} = \max(0, 2v - 1)$.
- Hardy model: logically contextual with $\text{CF} = 1/6$, and a *false negative* of $\mathbb{Z}$ -cohomology. The returned witness family uses a -1 coefficient and is verified compatible, which agrees with the note in `glue.py`.
- Mod-3 grid (Problem 13.11): $\gamma \neq 0$ at 9/9 over $\mathbb{Z}$ and $\mathbb{Z}/3$; 0/9 over $\mathbb{Q}$, $\mathbb{Z}/2$ and $\mathbb{Z}/5$. Both controls give 0; $\text{CF} = 1, 0, 0$; the fragility of Proposition 11.8 is reproduced exactly.

15 Problem 26: holonomy from estimated operations

Operations are observed through a *design*: integer functionals $a_k \in \mathbb{Z}^E$, with observations $y_k = a_k \cdot g + \epsilon_k$ and a noise law independent of g .

Proposition 15.1 (Ring-theoretic identifiability). *proved, computed* Let $A \in \{\mathbb{R}, \mathbb{R}/\mathbb{Z}, \mathbb{Z}/n\}$ with $R = \mathbb{Q}, \mathbb{Z}, \mathbb{Z}/n$ respectively. The holonomy $h(L) = \ell_L \cdot g$ is identifiable iff $\ell_L \in \text{span}_R\{a_k\}$.

Proof. If $\ell_L = \sum \lambda_k a_k$, then $h = \sum \lambda_k (a_k \cdot g)$. Conversely, A is an injective cogenerator over R . If $\ell_L \notin \text{span}$, there is g' with $a_k \cdot g' = 0$ for all k and $\ell_L \cdot g' \neq 0$, so g and $g + g'$ are observationally equivalent. $\square$

Observing only each loop traversed twice identifies h over $\mathbb{Q}$ and over $\mathbb{Z}/\text{odd}$, but not over $\mathbb{R}/\mathbb{Z}$ or $\mathbb{Z}/\text{even}$. Witness pairs are constructed in the tests.

Corollary 15.2 (T0 for operations). *proved*

- A design that reads each vertex through one route (a spanning-tree section) spans no cycle, so no loop is identifiable.
- An estimator that fits a field and sets $\hat{g} = \delta \hat{c}$ returns $\hat{h} \equiv 0$. Measured: 10^{-17} on a world with $h = 0.2$, while edge-local estimates give 0.205.

Proposition 15.3 (Calibrated test and sample cost). *measured; large-sample* The Wald statistic $T = \hat{h}^\top \Sigma^{-1} \hat{h}$, with $\Sigma = C \text{diag}(\text{se}^2) C^\top$, is $\chi_{\beta_1}^2$ under $[g] = 0$ when the circular-mean standard error is the delta-method one, $\text{var} = (1 - \rho_2)/(2m\rho_1^2)$.

- *Calibration: 4.2–6.4% false positives at nominal 5% for $\sigma \in [0.05, 0.35]$ turn ($m = 30$, 500 replicates). The naive standard error reaches 74%.*
- *Power follows the non-central χ^2 .*
- *Readings per edge for 80% power:*

$$N_{80} = \frac{|L|}{h^2} v(\sigma) c(\alpha, \text{power}, \beta_1), \quad v(\sigma) = \frac{(1 - e^{-8\pi^2\sigma^2})e^{4\pi^2\sigma^2}}{8\pi^2}.$$

This assumes equal per-edge noise and edge-disjoint loops. Measured power at the predicted N_{80} is 0.77–0.86.

- *The Cramér–Rao bound equals $e^{4\pi^2\sigma^2}/(8\pi^2)$ asymptotically, and the circular mean is 89–100% efficient, so the exponent is intrinsic.*

Proposition 15.4 (Discrete groups). *proved; measured looseness For $\mathbb{Z}/n$ with a symmetric channel, the mode estimator is exact with probability at least $1 - |E|(|A| - 1)(1 - (\sqrt{p_t} - \sqrt{p_w})^2)^m$. The bound holds but is about 10× loose (0.018 against a measured 0.001 at $m = 80$).*

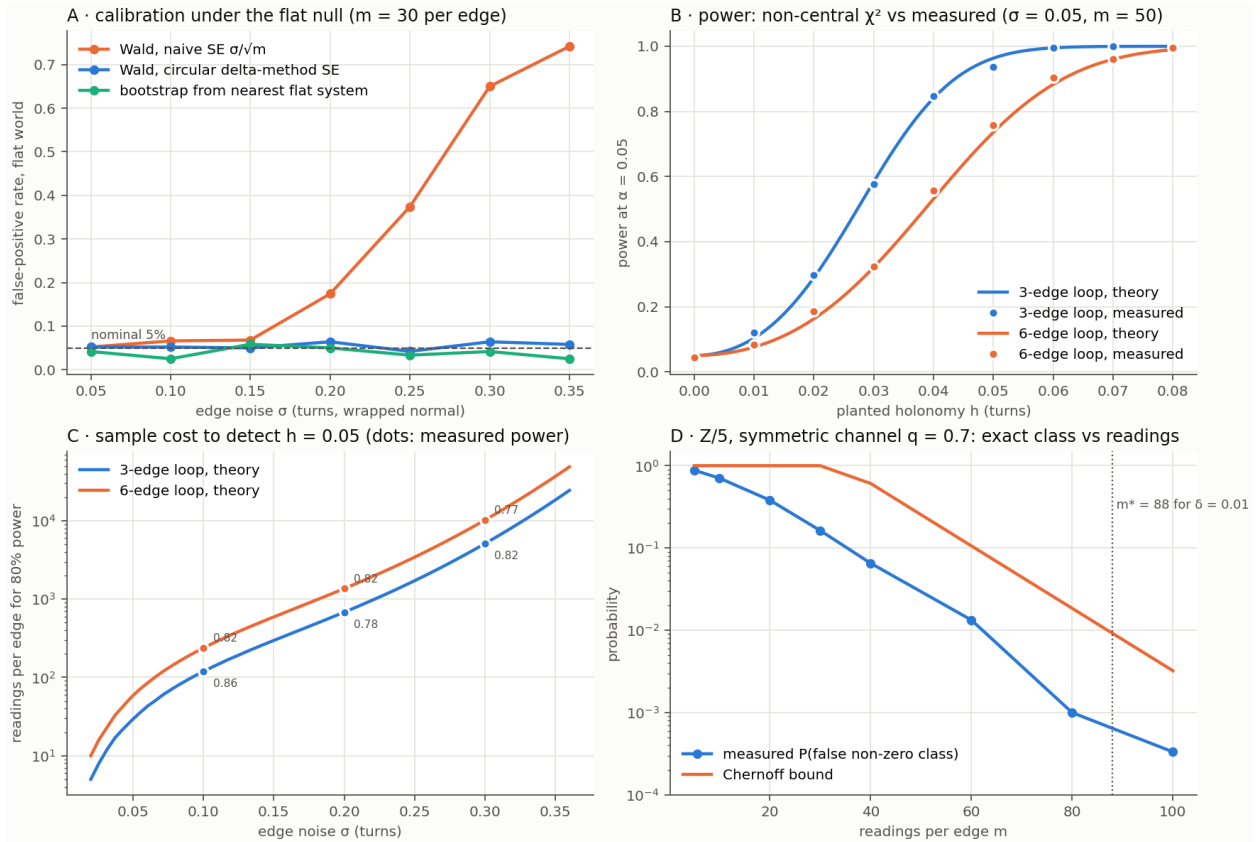


Figure 1: Problem 26 [S]. (A) calibration; (B) power against theory; (C) sample cost; (D) discrete class error against the Chernoff bound.

16 From operations to outcomes, around the probability layer

Theorem 16.1 (The operation-system model e_g). *proved, computed* For $(\Gamma, \mathbb{Z}/n, g)$, let the measurements be the vertex colours, the contexts the edges, and let e_{ab} be uniform on $\{(x, x + g_{ab})\}$. Then:

1. e_g is strongly contextual iff $[g] \neq 0$;
2. for $[g] \neq 0$, $\gamma(s) \neq 0$ for every section over every nonzero coefficient ring;
3. e_g is an instance of an all-versus-nothing model over $\mathbb{Z}/n$ [3].

Proof. (1) A global section is exactly a global field.

(2) Take a compatible R -family with $r_{C_0} = s$. Its restrictions to vertices are functions $f_v : \mathbb{Z}/n \rightarrow R$. Each support is the graph of a bijection, so compatibility forces $f_b(y) = f_a(y - g_{ab})$. Propagating the delta function $f = \delta_{x_0}$ from C_0 around a loop with $h \neq 0$ gives $\delta_{x_0} = \delta_{x_0+h}$, which is impossible when $R \neq 0$. $\square$

Proposition 16.2 (Layer 2 is blind to $[g]$). *proved* In e_g , every edge has $I(c_a; c_b) = \log n$ whatever g is. More generally, any per-context functional invariant under relabelling outcomes (every Shannon or Tsallis functional) is unchanged by post-composing an edge with a permutation.

The ring-invariance in Theorem 16.1(2) contrasts with the mod-3 grid, where γ vanishes over $\mathbb{Q}$. Ring dependence of γ is a property of the family, not of γ .

17 Lossy operations

Each edge carries a doubly stochastic channel K . Colours are uniform, so the Bayesian inverse is $K^\top$. The loop operator M_L is the product around the loop, and the edge-context model e_K has law $\frac{1}{n}K_{ab}$ on edge (a, b) .

Proposition 17.1 (Character eigenvalues). *proved* For covariant channels $C_{\mu_e} S_{g_e}$ on a finite abelian A , every character χ is an eigenvector of M_L with $\lambda_\chi = (\prod_e \hat{\mu}_e(\chi))\chi(h)$. For the noisy shift $K = (1 - \varepsilon)S_g + \varepsilon U$, $M_L = (1 - \varepsilon)^{|L|}S_h + [1 - (1 - \varepsilon)^{|L|}]U$ and $\lambda_\chi = (1 - \varepsilon)^{|L|}\chi(h)$.

Corollary 17.2. The labelled eigenvalue is base-point free, and its phase is $\chi(h)$ when every $\hat{\mu}_e(\chi)$ is real and positive. The unlabelled multiset is base-point free for any channels, since AB and BA share their nonzero spectrum, but on $\mathbb{Z}/p$ it sees only the order of h . Asymmetric noise adds its own phase: the readout is 2.285 instead of 2.

Proposition 17.3 (Information closed forms). *proved* $I(c_a; c_b) = \log_2 n + (1 - \varepsilon + \frac{\varepsilon}{n}) \log_2(1 - \varepsilon + \frac{\varepsilon}{n}) + (n - 1) \frac{\varepsilon}{n} \log_2 \frac{\varepsilon}{n}$. The loop-composite information is the same with $\varepsilon_L = 1 - (1 - \varepsilon)^{|L|}$. Both are independent of h .

Theorem 17.4 (Contextual fraction on a cycle). *proved* On a cycle of length L with noisy shifts and holonomy $h \neq 0$ in a finite abelian A ,

$$\text{CF}(e_K) = \max\left(0, 1 - \frac{L\varepsilon}{2}\right),$$

independent of $|A|$ and of h .

Proof. Upper bound on NCF. Weight each cell by its discrepancy $d = y - x - g$: $w(0) = 0$, $w(-h) = 1$, and $w = \frac{1}{2}$ otherwise. Discrepancies sum to $-h \neq 0$, so every colouring has weight at least 1. The dual value is $L \frac{\varepsilon}{n} (1 + \frac{n-2}{2}) = \frac{L\varepsilon}{2}$.

Lower bound (packing). Take colourings with one violation of discrepancy $-h$ (mass ε/n per edge), and colourings with two violations $(a, -h - a)$ (mass $\varepsilon/(2n(L-1))$ per ordered edge pair), spread over translations. These fill every off-graph class exactly, and the on-graph load $\frac{L\varepsilon}{2} - \varepsilon(1 - \frac{1}{n})$ fits within the capacity $1 - \varepsilon(1 - \frac{1}{n})$ iff $L\varepsilon \leq 2$.

Beyond the threshold. The model is affine in ε and non-contextual at $\varepsilon = 2/L$ and at $\varepsilon = 1$. $\square$

Theorem 17.5 (Any graph, lower bound). *proved* $\text{CF}(e_K) \geq \max(0, 1 - \ell_{\min}\varepsilon/2)$, where $\ell_{\min}$ is the shortest simple cycle with nonzero holonomy.

Observation 17.6. *measured* Equality holds in 44 configurations: θ -graphs, K_4 , shared edges, shared vertices, disjoint non-flat loops, and $\varepsilon \leq 0.8$. Only $\ell_{\min}$ matters, not the cycle basis. The general packing is open.

Counterexample 17.7 (What does not hold). *computed*

- *Information does not determine CF.* A 3-cycle and a 5-cycle over $\mathbb{Z}/3$ at $\varepsilon = 0.2$ have the same per-edge information, but $\text{CF} = 0.70$ and 0.50 .
- *Non-covariant noise is contextual on its own.* Flat systems reach CF up to 0.24 , and the phase readout of $h = 2$ spreads over $[1.1, 3.1]$.
- *Entropy is not a function of Fourier moduli.* A homometric pair has identical $|\hat{p}|$ but $H = 2.7714$ against 2.7689 bits.

layer	invariant	response to ε	detects h until
1	labelled eigenvalue of M_L	modulus $(1 - \varepsilon)^{ L }$, phase fixed	$\varepsilon \rightarrow 1$
2	per-edge and loop I	decreases, independent of h	never
3	CF	$1 - \ell_{\min}\varepsilon/2$	$\varepsilon = 2/\ell_{\min}$
3	γ	support becomes full	$\varepsilon = 0$ only

Table 9: The same noise acting on four invariants [S].

Proposition 17.8 (Orientation for $\mathbb{Z}/3$ and $\mathbb{Z}/5$). *proved, computed* For odd primes, $h \neq -h$ whenever $h \neq 0$, so the labelled phase always detects orientation. Reversing the loop inverts the readout ($1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ on $\mathbb{Z}/5$). The unlabelled multiset and CF do not detect orientation.

- On $\mathbb{Z}/3$ a non-flat holonomy is exactly a chirality, read from the sign of $\text{Im } \lambda_1$.
- On $\mathbb{Z}/5$ the phase splits into a magnitude class, $\text{Re } \lambda_1/|\lambda_1| = 0.309$ for ± 1 and -0.809 for ± 2 , and an orientation.
- The magnitude class is meaningful only with fixed state labels, since $x \mapsto 2x$ carries $h = 1$ to $h = 2$.

18 The support dial: Tsallis entropies

Proposition 18.1. *computed; the cocycle property is Vigneaux's* Let $S_\alpha = (1 - \sum p^\alpha)/(\alpha - 1)$ with the deformed action $(X \cdot_\alpha F)(P) = \sum_x P(X=x)^\alpha F(P \mid X=x)$.

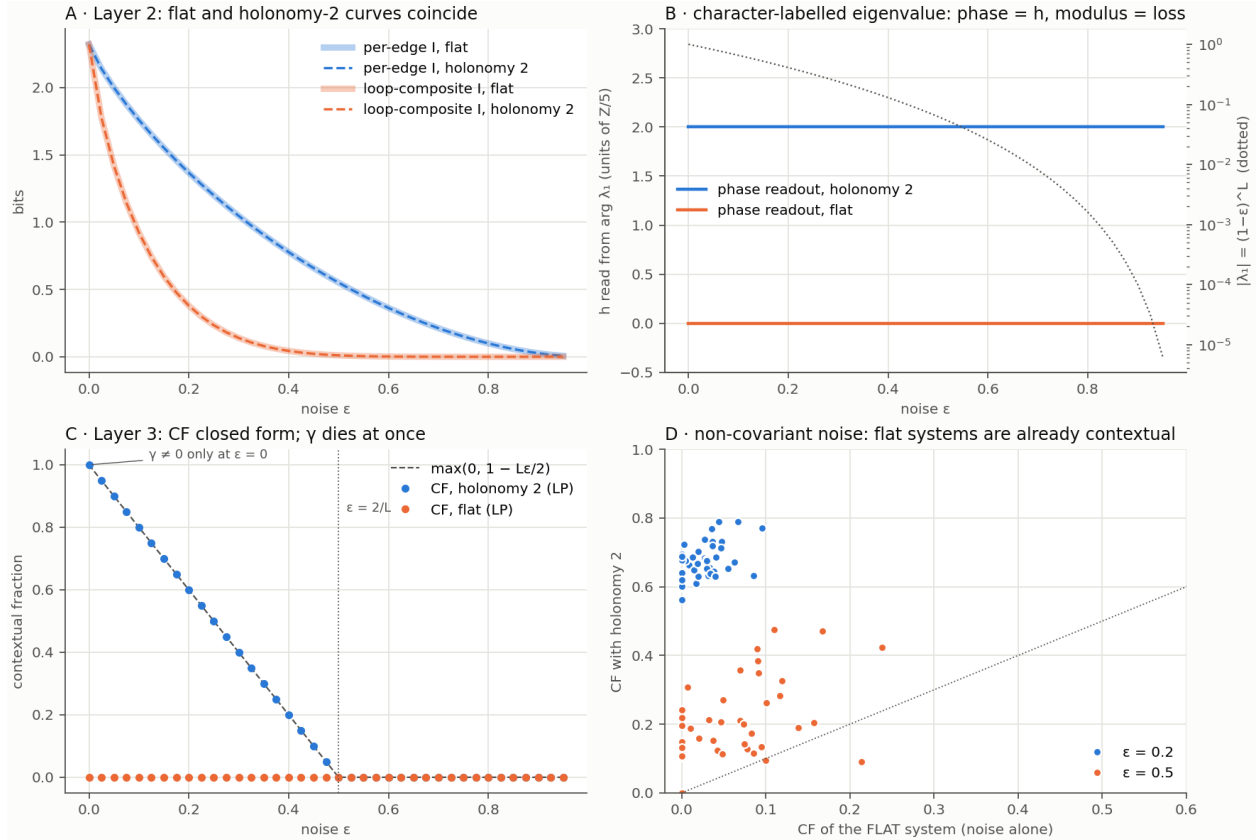


Figure 2: Noisy operations on a 4-cycle over $\mathbb{Z}/5$, with $h = 2$ against a flat system [S].

- S_α is a δ_α -cocycle (defect below 3×10^{-15}), but not a cocycle for Shannon's action (defect up to 1.8).
- $S_0 = |\text{supp}| - 1$, and $S_\alpha \rightarrow H$ as $\alpha \rightarrow 1$.
- Mass ϵ added to an empty cell changes S_0 by 1, and S_α by $\sim \epsilon^\alpha$.
- $S(X) + S(Y) - S(XY)$ reaches -1.71 at $\alpha = 0.5$.

α therefore interpolates between AMB's possibilistic lens ($\alpha = 0$) and Shannon ($\alpha = 1$). Every S_α is symmetric, so Proposition 16.2 still holds.

19 The decorated Bregman–Čech nerve

Tokens are points $p_i \in \Delta(\text{states})$ with a time τ_i . The radius of a simplex is $\min_c \max_i D_{\text{KL}}(p_i \| c)$.

Lemma 19.1. *proved, computed* The balls $\{p : D_{\text{KL}}(p \| c) \leq r\}$ are convex, since KL is convex in its first argument, so the nerve lemma applies. For a pair, the optimum lies on the segment between the two points; it is found by bisection and certified by the minimax dual $\max_\lambda [\lambda D_{\text{KL}}(p \| m) + (1 - \lambda) D_{\text{KL}}(q \| m)]$ to 10^{-9} .

Construction 19.2 (Decoration). • Edges are oriented by time (“A before B”) and labelled by the best-aligning group element, with its margin over the runner-up.

- Triangles carry their boundary holonomy; a nonzero value is a non-flat event.

- Each H_1 bar is labelled by the holonomy of its birth cycle.

Proposition 19.3 (Invariants for $\text{Aff}(\mathbb{Z}/p)$). *proved* Loop labels depend on the base point only by conjugation. The conjugacy classes are: the identity; all nontrivial translations, as one class; and one class per multiplier $u \neq 1$. For odd p , $\text{Aff}(\mathbb{Z}/p)^{ab} = (\mathbb{Z}/p)^\times$, so only u is a function of the H_1 class. Relabelling tokens preserves the class and not the element (tested).

Proposition 19.4 (Walls). *proved; consequence measured* A permutation σ of n states fixes a face of codimension $n - \#\text{cycles}(\sigma)$. In $\text{Aff}(\mathbb{Z}/3) \cong S_3$ the reflections are transpositions, fixing codimension-1 walls. The quotient is a Coxeter chamber, and every planted holonomy is absorbed into a flat labelling (0/100 recovered). Every non-identity element of $\text{Aff}(\mathbb{Z}/5)$ has codimension ≥ 2 .

Observation 19.5. *measured* On $\text{Aff}(\mathbb{Z}/5)$, rings with every edge margin $\geq 10^{-3}$ are recovered in 254/255 cases; rings with an ambiguous edge are wrong in 57/145. On $\mathbb{Z}/3$ translations, both orientations are recovered and reversing time inverts the label.

Limits *measured*:

- Dirichlet noise on tokens (concentration 2000 $\rightarrow$ 200) raises ring-label error from 15% to 48–63%.
- The $\mathbb{Z}/2$ barcode readout is confounded by shortcut loops of the quotient and by coefficients: 18/20 correct on $\mathbb{Z}/5$ translations and 8/12 on $\mathbb{Z}/3$.
- In the non-abelian case, bars with non-identity labels can die at flat triangles.

Semantics adopted. Negation is *partial*: it is undefined on U and sends U to an explicit *not-applicable* state. Operations therefore form an inverse monoid, not a group, and are lossy.

20 Two facets: restriction and extension

Let $i : \Sigma \rightarrow \mathcal{S}$ include the cover into the information site.

Proposition 20.1. *proved, computed*

1. Restriction $R: P \mapsto (\pi_C P)_C$ has image exactly the non-contextual models.
2. Extension $E: e \mapsto \text{Ext}(e) = \{P : R(P) = e\}$ is a convex polytope, nonempty iff $\text{CF}(e) = 0$. If $\text{CF} = 0$, there is $b \geq 0$ with $\sum b = 1$ and $R(b) \leq e$, and equal context totals force $R(b) = e$.
3. $P \in E(R(P))$, and $R(E(e)) = \{e\}$ whenever $E(e) \neq \emptyset$.
4. The canonical member of $E(e)$ is the maximum-entropy extension, found by iterative proportional fitting (Csiszár). IPF fails to converge on contextual models.

Observation 20.2 (Graded extension). *measured* The optimal non-contextual parts, of mass $1 - \text{CF}$, form a face. In CHSH it is a single point for every contextual model tested (39/39); in the three-party GHZ scenario it never is (0/10). The maximum-entropy point of the face gives a unique representative.

These maps act on models, not on cohomology, so they are consistent with Theorem 13.1. The attached drafts' forward map, which sent γ/CF to the spectral modulus, is not a functor.

21 Not-applicable entries: the gated test

Records are windows of labels (T, F, could-T, could-F, U, n/a), with a *mask* marking variables that are not in the context. The mask is distinct from the *n/a value*.

- Construction 21.1** (Pipeline). 1. GYO reduction of the maximal observed contexts. If the cover is acyclic, the pipeline stops (Theorem 10.4); any $CF > 0$ is then inconsistency, not contextuality.
2. Possibilistic test across support thresholds, with its null band.
3. CF by LP, with the signalling defect reported beside it.
4. Null: each record keeps its mask, and its values are redrawn from the maximum-likelihood consistent law. That law is found by missing-data EM (Theorem 10.2) and pinned to maximum entropy.

stream	gate	CF	verdict
random N/A, complete records present	acyclic	—	stop: the AMB half is empty
honest, one N/A per record	cyclic	0.029	null mean 0.027, $p = 0.39$: stop
structured, shift twist	cyclic	0.858	above all 30 null replicates (max 0.044)
structured, negation with $U \rightarrow n/a$	cyclic	0.859	above the null, but signalling 0.18

Table 10: The gated test on 9,000-record streams [S].

Proposition 21.2 (The deficit). *proved (zero set); measured (the rest)* The deficit is the per-record log-likelihood lost by the best consistent law, $\min_P \sum_C w_C D_{KL}(e_C \| P_C)$.

- Zero set. For no-signalling tables it vanishes iff $CF = 0$: the minimum is attained, and KL is zero iff the tables coincide.
- Not a function of CF. At $CF = 0.5$ on noisy-shift cycles it is 0.101, 0.068, 0.051 for $L = 3, 4, 5$.
- Closed form at $\varepsilon = 0$: it equals $\log(L/(L-1))$ (measured to 10^{-4}).
- Sampling floor. On honest data it falls as about $1/N$ (0.0079, 0.0010, 0.00015 for $N = 10^3, 9 \cdot 10^3, 8.1 \cdot 10^4$); on contextual data it stays flat at about 0.282.
- Bootstrap intervals: [0.270, 0.287] for the shift twist and [0.291, 0.309] for the negation twist.

Observation 21.3 (The ridge). *measured* When contexts are pairs, the three-way interaction is unidentified. EM from different starts reaches the same likelihood with laws 0.005–0.023 apart in L_1 . This is the upper-level identifiability failure of Part I in three variables. The maximum-entropy pin removes the start dependence (tested); without it, a null on larger contexts would measure ridge drift.

Observation 21.4 (Thresholded γ). *measured* Near a cell’s expected mass, sampling flips cells in and out of the support and manufactures logical witnesses. The null fires in 20/20 replicates at threshold 0.005 on honest data, and at 0.1 on the contextual streams’ null. γ is admissible only where the null band is near zero: at 0.02 and 0.15, both structured streams give $\gamma \neq 0$ at 5/5 against a null of 0/20. Report the full sweep.

Part IV

Status

22 Status of every claim

- **Proved:**

- T0 and its corollary; holonomy obstruction and connecting class;
- Propositions 8–10 of [O]; non-unification; vanishing (trivial action); Corollary 28;
- the p -adic proposition;
- identifiability (Prop. 15.1) and its T0 corollary; the Chernoff bound;
- Theorem 16.1 and Layer-2 blindness; character eigenvalues;
- Theorem 17.4; the lower bound of Theorem 17.5;
- orientation on $\mathbb{Z}/3$ and $\mathbb{Z}/5$; convexity of KL balls and exact pair radii;
- $\text{Aff}(\mathbb{Z}/p)$ invariants; the codimension formula;
- R and E ; the zero set of the deficit.

- **Computed:**

- V0–V4a identities; H^0 , H^1 , H^2 of small semilattices;
- the mod-3 grid over five rings; possibilistic fragility; Hardy false negative;
- Baudot–Bennequin identities; the Tsallis δ_α cocycle;
- Counterexample 17.7.

- **Measured:**

- transformer negative; split–merge recovery; instrument calibrations;
- Wald calibration and N_{80} ; equality in Theorem 17.5;
- wall absorption; $\text{Aff}(\mathbb{Z}/5)$ recovery; graded-extension uniqueness;
- the gated-test table; deficit behaviour; ridge; threshold windows.

- **Open:**

- O1 identifiability threshold $\theta(V, L, \ell, m)$; O2 minimal sufficiency of merges; O4 power of β_1 ; O6 split–merge at $V = 1024$, $L = 16$; O7 recurrent-depth transformers;
- Problems 21, 22 (23 blocked; 24 settled);
- $H^{\geq 2}$ for δ and δ_α ;
- the packing proof for Theorem 17.5; an invariant separating noise from holonomy under non-covariant noise;
- integer-coefficient persistence for decorated barcodes;
- the conjecture that operations visible to Layer 2 are lossy and admit no *cohomological* invariant. It is refuted in part: the loop spectrum is a non-cohomological invariant;
- extending the gated test to triple contexts.

- **Withdrawn or corrected:**

- the embedding probe [H];
- “ $I_2 = \delta H$ ”;
- the two-point entropy formula for noisy shifts;
- “information bounds dictate CF”;
- “the spectrum reads h ”: it is the labelled eigenvalue;
- “the optimal non-contextual part is generally not unique”: it depends on the scenario;
- “deficit independent of CF”: it was then established by the sweep;
- a claim in the nerve module that non-identity bars die only at non-flat events: it fails for $\mathbb{Z}/2$ coefficients and non-abelian groups;
- the “BBV” expansion as a Bell-nonlocality framework: no such framework could be identified.

23 Method notes

- Every detector needs a planted positive and a null. The engine’s T0 bug and the thresholded- γ false positives were caught this way.
- Use Wasserstein, not bottleneck, distance for near-degenerate diagrams [H].
- Plug-in MI bias is $(K - 1)(L - 1)/2N$; bootstrap nulls cancel it [H].
- Read the proposition, not the summary.

24 Reproduction

- [H]: the command appendix of the hierarchy paper (`build_corpus_syn.py`, `splitmerge.py`, `estimator4.py`, `estimator6.py`).
- [O]: `holonomy.py`, `amb_gamma.py`, `parity_glue.py`, `h2_rational.py`, `h2_integral.py`.
- [S]: `python -m pytest (107 tests), examples/holonomy_estimated.py, examples/channels.py`, and modules `outcome`, `holonomy`, `channels`, `nerve`, `functors`, `na_contexts`.

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